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Project Specifications and Risk Assessment
Low-cost Self-sustainable Deployable Network

Group 007

(names removed)

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This is an excellent report, scoring 50 / 50.

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1 High Level Description

1.1 Motivation

Every year, the Canadian telecommunications industry spends \$9.2 billion on planning and development [1] to provide Internet connectivity across the country. In mature countries like Canada, wide-spread Internet availability accounted for over 20% of gross domestic product (GDP) growth in the past 5 years [2]. However, 15% of rural Canadians are still being left without Internet connection [1], and for developing countries lacking capital expenditure, Internet has become an impossibility for 68% of households [3]. Internet dead-zones can also occur beyond rural and developing regions; efforts to restore Internet connectivity to areas affected by disaster, such as the 2015 Nepal Earthquake region, can take well over a month [4].

The primary issue is that traditional wireless infrastructure lacks the flexibility to economically extend coverage to these last-mile regions. Typical base-stations not only require readily available power sources, but also must be connected by fiber optic cables, which need land permits and expensive construction. An arduous and time-consuming planning phase would also be necessary to ensure that the base-stations avoid interference, successfully communicate with each other, and provide adequate range. The immense economic and time costs associated in wireless infrastructure development has greatly reduced Internet availability in locations where it is needed most; rural, developing, and disaster-stricken regions.

1.2 Project Objective

The objective is to design a low-cost solution that can be quickly and easily deployed to provide Internet access in regions with little or no existing infrastructure.

1.3 Block Diagram

The proposed solution is a small-scale wireless base-station (node) that is uniquely designed to be powered solely from solar energy and sustain a computer network autonomously. The architecture can be shown in **Figure 1**, and is composed of 3 main subsystems: communications, network and microcontroller, and power.

1.3.1 Communications Subsystem

Network interface controllers (NICs) are components that can provide wireless connectivity such as Wi-Fi adapters. A total of two are required; one for connections between other base-stations (back-bone connection), and one for connections between users or clients of the network (access point connection). Both require antennas, but for this particular design, the back-bone connection is prioritized and a fully designed switched antenna array will be built from the ground up. Each antenna array element can be selected to provide a beam in

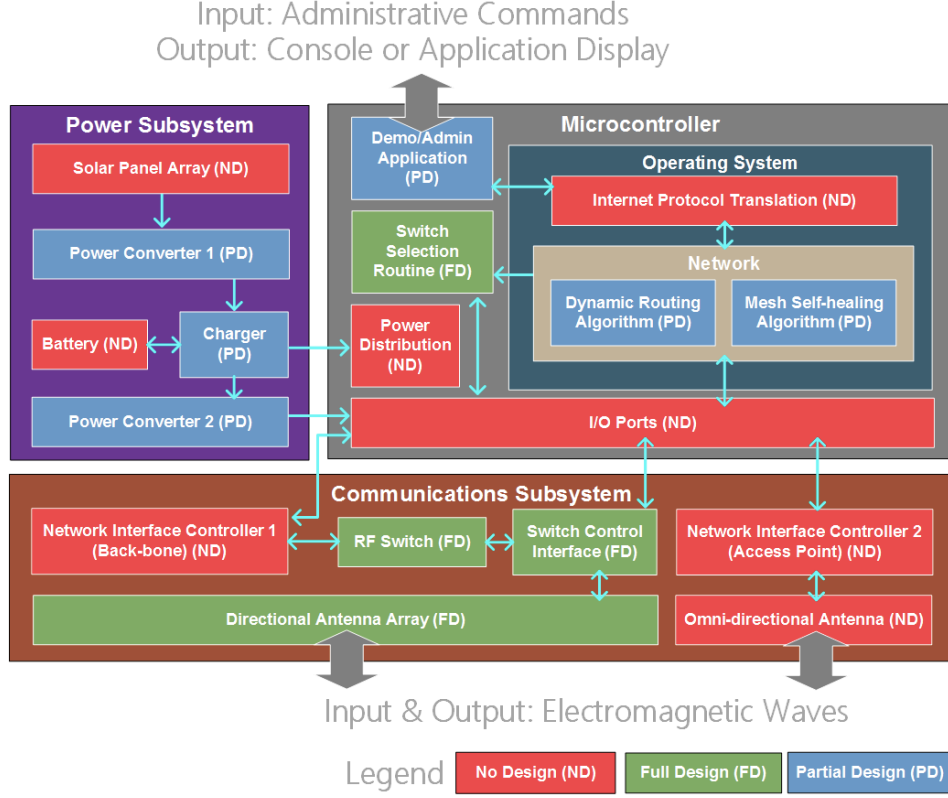


Figure 1: Block diagram of the low-power base-station indicating subsystems and connections between each element

a specific direction. The "RF switch" block includes the design of a matching network to maximize power to the antennas, and is controlled by the "Switch Control Interface" which includes digital circuits that interface to the microcontroller.

1.3.2 Network and Microcontroller

The microcontroller runs a mesh network protocol that includes two main algorithms; dynamic routing and mesh self-healing. The dynamic routing algorithm determines which base-station to transmit to. The mesh self-healing algorithm dictates the response when nearby base-stations approach or leave the network. Existing implementations of these algorithms exist, but careful selection and tweaking may be necessary as part of the network design. The dynamic routing algorithm will be used by the switch selection routine to determine which switch to activate. In addition, tools for administration, debugging, and demonstration of the network will be partially designed from existing applications.

1.3.3 Power Subsystem

The main function of the power subsystem is to provide constant power to the microcontroller and NICs from a solar panel array and a lithium ion battery. In order to do so, two power converters and a lithium-ion charger are necessary. These components are based on existing integrated circuits, but the selection and integration of the components onto a single PCB require design insight.

2 Project Specifications

The project specifications can be sub-divided into functional specifications, and non-functional specifications. Each is further sub-categorized as essential or non-essential given how important the specification is in accomplishing the project objective.

2.1 Functional Specifications

Seven essential functional specifications were identified and can be found in **Table 1**. Similarly, two non-essential functional specifications can be found in **Table 2**.

2.2 Non-functional Specifications

Six essential non-functional specifications can be found in **Table 3**, while the three non-essential non-functional specifications can be found in **Table 4**.

3 Risk Assessment

The major risks identified are detailed in **Table 5** and include hardware acquisition, technical risks, and group member availability and competency.

Table 1: Functional Essential Specifications

Specification	Location	Description
Back-bone range	Communication subsystem	Base-stations must be able to connect to an existing network within 500 m radius of the base-station in open space
Maximum Stable Throughput	Network & Microcontroller	Maximum throughput of a single link between two base-stations must exceed 512 kbps for over 1 minute anywhere within back-bone range under line-of-sight. This value may vary outside the above stated conditions.
Packet loss	Network & Microcontroller	Average packet loss percentage of a single link between two base-stations at maximum stable throughput must be less than 5%
Automatic network configuration	Network & Microcontroller	Base-stations can automatically join an existing network within 1 minute of entering back-bone range
Self healing	Network & Microcontroller	The remaining 2 base-stations of a 3-node chain can automatically connect to each other within 1 minute after removing the middle base-station
Regulatory restrictions	Communication subsystem	Antenna gain and transmission power must not exceed Canadian regulatory power restrictions
Latency	Network & Microcontroller	Maximum latency of any single link must be less than 50 ms

Table 2: Functional Non-Essential Specifications

Access point range	Communication subsystem	Clients must be able to connect to a base-station within 250 m radius of the client in open space
Scalability	Network & Microcontroller	The maximum throughput after adding an additional base-station is greater than 75% of maximum stable throughput
Administration	Network & Microcontroller	Network can be accessed remotely using a web-interface and can perform management operations such as modifying security and authentication settings

Table 3: Non-functional Essential Specifications

Specification	Location	Description
Maximum power dissipation	Overall system	Maximum power dissipation from the system while transmitting at its maximum rate is below 1.4 W
Sustainability	Overall system	System should be able to sustain itself on battery power while transmitting at maximum rate for at least 12 hours
Power output	Power subsystem	Charger and power converters must be able to supply maximum power dissipation at the operating voltage range of the microcontroller and NICs
Simultaneous charge and use	Power subsystem	Battery can be charged while base-station is fully functional
Conversion efficiency	Power subsystem	Total efficiency of converting solar energy to usable electrical energy for the microcontroller and NICs is greater than 15%
Self Recovery	Power subsystem	An unpowered base-station should be able to automatically power on when battery capacity reaches above a certain threshold

Table 4: Non-functional Non-Essential Specifications

Specification	Location	Description
Ruggedness	Overall system	System can be dropped from 1m height with no effect on throughput of ongoing connections
Low cost	Overall system	Total cost of each base-station is less than \$300
Portability	Overall system	The device weighs less than 2 kg and no more than 50 cm x 50 cm x 40 cm (W x L x H) in size

Table 5: Risk impact and probability identification with possible solutions to mitigate the risk

1	Hardware Acquisition
	Probability: High
	Impact: Medium
	The group is ordering hardware including, but not limited to, solar panels, power converters, antennas, switches and integrated circuits from electronic device distributors. Shipping items from domestic and international sources carry risks such as delays, damages and wrong addressing. Should any of these risks come to fruition, the project time line may be delayed.
	Solution
	As domestic shipping is both more faster and reliable than international shipping, domestic suppliers such as DigiKey Canada would be given preference when choosing between competing suppliers. Should a component be exclusively available from international suppliers, we can prepare for the risk of shipping delays by acquiring a similar part that is more readily available. For example, a 9 W solar panel with specific maximum power point voltage ordered from Europe is delayed. We can find a solar panel with a lower power rating that is easier to get a hold of. This solar panel may not have the exact specifications that is required, but it can provide a proof of concept while shipping issues are being resolved. Hardware should be ordered as soon as possible when no other blocking requirements are in place. Expedited shipping options will be taken into consideration and used if significant time reductions can be achieved.
2	Technical Risks
	Probability: Medium
	Impact: High
	Due to the novelty of the project and the some of the components being used, there are risks in not being able to fully predict limits in network performance and the challenges in implementing the design. For example, while there are published results detailing how a similar network will scale to a small (less than 30) amount of nodes, there are very few for large scale implementations which may limit the future viability of the project. Furthermore, the planned low-power microcontroller component is new and has less developer support than other more popular and established microcontrollers. This can lead to more effort being needed to debug the component, leaving less time to focus on other parts of the design.

Solution	
A simulation can be built before the dedication of resources that can help to predict the scaling capability of the network. The simulation would represent the ideal case and provide an upper-bound for expected performance. Having alternate components that can be used instead of the selected component in situations where technical limitations arise can reduce the time needed to circumvent barriers in component support.	
3	Group member availability and competency
Probability: Medium	
Impact: High	
Group members may not have all the necessary skills or experience to fulfill their respective duties. This can cause delays in the project time line as some group members take time to get themselves up to speed. In addition, the group members are enrolled in 4A electrical and computer engineering courses. This may cause schedule conflicts where it may be difficult for the entire group to meet at once throughout the weekdays. Coursework and job interviews may also be prioritized by group members over the completion of the project, which can result in reduced dedication to finishing project goals and a lower quality of work.	
Solution	
To combat scheduling conflicts, group members can split up their meeting times throughout the weekdays. For example, breaking up the group into meetings of 2 or 3 members for a meeting is much easier than attempting to group all 5 members at once. The group can then follow up over the weekend when all 5 members can be present. A set schedule must be followed with minimum work time to allow the project work to coexist with the coursework and interviews. In order to combat familiarity issues with certain technical topics, the team members can schedule meetings with lab instructors and professors with expertise in the technical area.	

References

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